Rishika Bera

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Portfolio: rishika2024.github.io | Citizenship: United States

EDUCATION

Northwestern University Evanston, IL
MS in Robotics Sep 2025 - Aug 2026
Robotic Manipulation, System Dynamics, SLAM, Deep Learning, Embedded AI, Generative AI

Indian Institute of Technology - Jodhpur Rajasthan, India
B.Tech in Mechanical Engineering Dec 2021 – May 2025
Classical Control Theory, Kinematics & Dynamics, Vibrations, Thermodynamics, Fluid Mechanics

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, HTML, CNC G-Code

Robotics: ROS 2, Robot Driver Development, UAV, SLAM, Robot Kinematics & Dynamics, Path Planning, Control Systems, PID Control, Closed-Loop Feedback, Perception, Sensor Fusion, Motion Planning

Machine Learning & AI: Machine Learning, Deep Learning (MLPs, CNNs, Autoencoders, GANs, Transformers, Diffusion Models), Reinforcement Learning, LLMs, PyTorch

Software & Libraries: PX4, MoveIt, OpenCV, Gazebo, RViz, CoppeliaSim, Unity3D, Simulink

Tools: Linux, Git, SSH

Hardware: Meca500 Arm, Pixhawk, Franka Arm, Interbotix Arm, RealSense Camera, Arduino, Raspberry Pi, Encoders

Design & Manufacturing: SolidWorks, Onshape, Blender, 3D Printing, CNC Machine, Soldering, Wiring

WORK EXPERIENCE

Purdue University May 2024 - Jul 2024

Summer Undergraduate Research (SURF) Intern

- Built a multimodal pipeline that scrapes, categorizes, and filters text, image, and video data using ChatGPT API for relevance scoring
- Developed a video segmentation module to extract relevant clips and filter irrelevant content
- Implemented LLM driven image attribute extraction achieving greater than 96% classification accuracy

PROJECTS

Robotic 3D Printing - Meca500 | (ROS 2, MoveIt 2, Python, C++, G-Code, Motion Planning) Apr 2026 - Jul 2026

- Built a robotic 3D printing system using a Meca500 6-axis arm with a Creality Ender 3 extruder, enabling printing on surfaces unreachable by standard gantry printers
- Built a ROS 2 hardware interface bridging the Meca500 vendor API to MoveIt 2
- Developed a G-Code to end effector waypoint parsing pipeline
- Used Pilz Industrial Motion Planner for linear and arc motions, OMPL with RRT for collision-free path planning to print bed
- Validated by printing a 50mm cube on hardware achieving 0.22mm and 0.02mm error on X and Y axes

RL for Embedded 3D Print Path Optimization | (ROS 2, MoveIt 2, Python, C++, RL) Jul 2026 - Present

- Developing an RL policy to optimize print paths for embedded 3D printing of polymer lattices using 6DOF joint-space motion
- Training the agent to avoid collisions with printed structures and maintain correct extruder angle
- Simulating and visualizing the full print process using MoveIt 2

EKF SLAM with TurtleBot3 | (C++, ROS 2, SLAM, Kalman Filter, LiDAR, Wheel Odometry) Jan 2026 - Mar 2026

- Developed a full SLAM pipeline from scratch in C++ with an Extended Kalman Filter, performing state prediction and measurement updates
- Fused wheel-odometry and LiDAR measurements for robot pose and landmark state estimation
- Extracted geometric landmarks from 2D LiDAR scans via point-cluster segmentation and circle-fitting
- Used Mahalanobis distance thresholding for unknown data association
- Wrote unit and integration tests in C++ (catch_ros2) and validated the full pipeline in RViz on a simulated TurtleBot3

Bug-Sorting - Franka Robot Arm (Group Project) | (Python, ROS 2, OpenCV) Dec 2025

- Contributed to developing an autonomous system to catch and sort moving HexBugs with the Franka Arm
- Built a vision system using OpenCV and RealSense to detect, label, and track HexBugs in real-time
- Explored continuous tracking and pick-and-place motion planning using ROS 2 and MoveIt

Morphing Aerial–Ground Transformer Robot | (CAD, Raspberry Pi, PX4, Pixhawk, UAVs) Jan 2026 - Apr 2026

- Designed a hybrid aerial-ground robot in CAD, featuring a hinge mechanism and planetary gear wheel system
- Assembled and flight-tested a quadrotor on an F450 frame, refining 3D-printed parts across multiple revisions
- Integrated PX4 with Raspberry Pi for teleoperated mode-switching between aerial and ground operation